

PROGRESS REPORT WEEK 3
PARALLEL PASSWORD HASHING (CPU & GPU)



DOSEN PENGAMPU :
SEPTIAN GEGES, S.KOM., M.KOM.

DISUSUN OLEH :
KELOMPOK 1

KELVIN ANDRE HARDIAN	(2430305030011)
MUHAMMAD SHAUFI Wafa B.S	(2430205030028)
GIOVANNI MARANATHA	(2430305030022)
FAJAR OKTAVIAN RAMADHAN	(2430205030023)

JURUSAN TEKNIK INFORMATIKA
FAKULTAS TEKNIK
UNIVERSITAS PALANGKA RAYA

2026

1. STATUS MODUL

Modul	Status	Keterangan
main.py	Selesai	Menjalankan CPU & GPU benchmark
cpu.py	Selesai	CPU brute force + multithreading
gpu.py	Selesai	GPU brute force menggunakan CUDA
hashing.py	Selesai	Implementasi MD5 hashing
generator.py	Selesai	Generate kandidat password di GPU
comparator.py	Selesai	Membandingkan hash
tests/	Selesai	Unit testing menggunakan pytest
README.md	3. KENDALA Selesai	Dokumentasi project

2. HASIL IMPLEMENTASI

CPU

- Sequential baseline berhasil dijalankan
- Multithreading berhasil ditambahkan menggunakan ThreadPoolExecutor
- Jumlah thread dapat dikonfigurasi

GPU

- CUDA berhasil dijalankan menggunakan Numba
- Candidate generation berjalan paralel di GPU
- GPU berhasil menemukan password berdasarkan hash target

Testing

- Unit testing menggunakan pytest berhasil dijalankan
- Output CPU dan GPU menghasilkan hasil yang sama

3. KENDALA

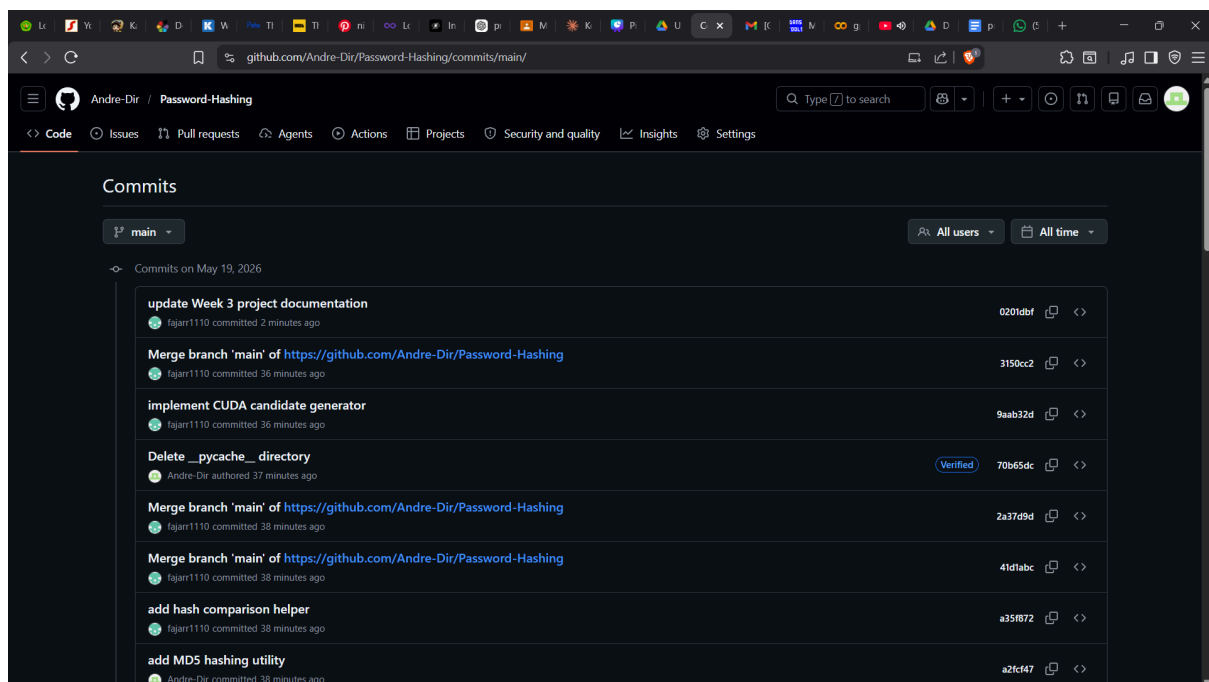
Kendala	Solusi
CUDA lokal gagal	Menggunakan Google Colab
Python 3.14 tidak kompatibel	Menggunakan Python compatible version
Error nvvm.dll	Menggunakan environment Colab
GPU underutilization warning	Menyesuaikan jumlah thread dan block

4. GIT WORKFLOW

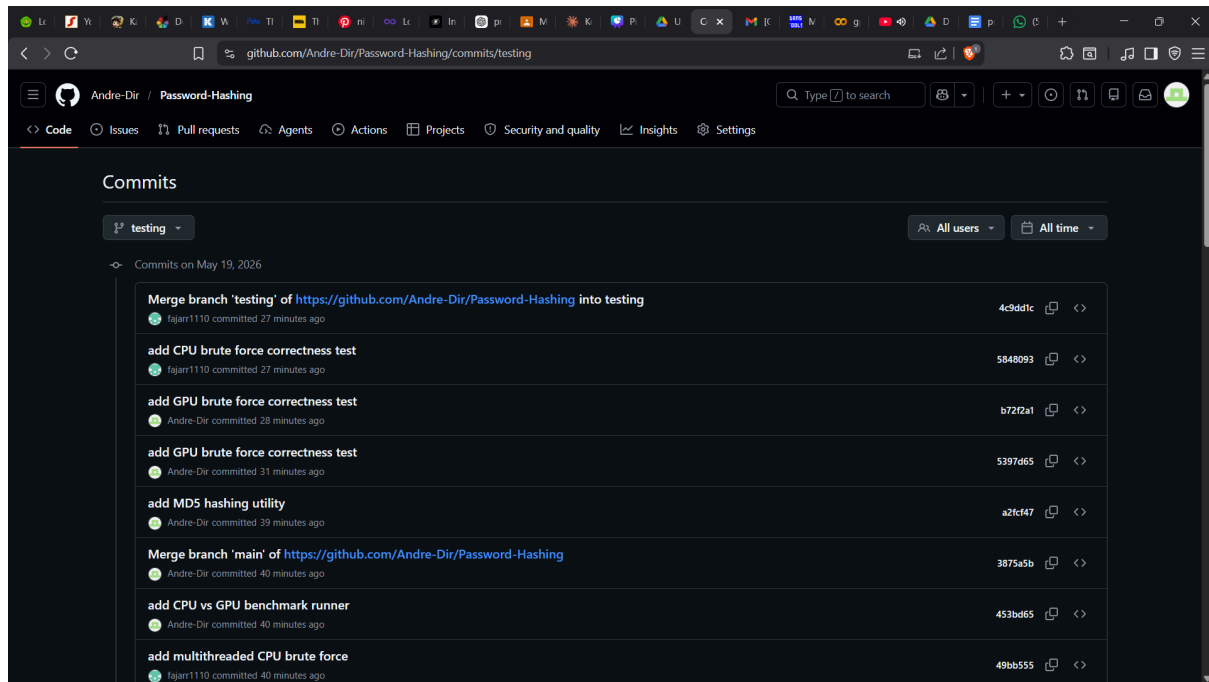
Branch yang digunakan:

- main
- testing

Testing dipisahkan dari implementasi utama agar kode stabil tetap aman.



Gambar 1 Commit di Branch Main



Gambar 2 Commit di Branch Testing

5. RENCANA WEEK 4

- Benchmark lebih detail CPU vs GPU
- Menambahkan grafik performa
- Analisis speedup
- Optimasi CUDA kernel
- Finalisasi laporan dan presentasi

6. LAMPIRAN

```

=== BENCHMARK ===
CPU time : 0.3874s
GPU time : 4.1884s
Speedup : 0.09x

```

Gambar 3 Hasil Benchmark CPU dan GPU

```

===== test session starts =====
platform linux -- Python 3.12.13, pytest-8.4.2, pluggy-1.6.0
rootdir: /content
plugins: anyio-4.13.0, langsmith-0.7.34, typeguard-4.5.1
collected 3 items

test_cpu.py . [ 33%]
test_gpu.py . [ 66%]
test_hashing.py . [100%]

===== warnings summary =====
test_gpu.py::test_gpu_found
/usr/local/lib/python3.12/dist-packages/numba_cuda/numba/cuda/dispatcher.py:696: NumbaPerformanceWarning: Grid size 1 will likely
warn(errors.NumbaPerformanceWarning(msg))

test_gpu.py::test_gpu_found
/usr/local/lib/python3.12/dist-packages/numba_cuda/numba/cuda/dispatcher.py:696: NumbaPerformanceWarning: Grid size 3 will likely
warn(errors.NumbaPerformanceWarning(msg))

test_gpu.py::test_gpu_found
/usr/local/lib/python3.12/dist-packages/numba_cuda/numba/cuda/dispatcher.py:696: NumbaPerformanceWarning: Grid size 69 will likely
warn(errors.NumbaPerformanceWarning(msg))

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 3 passed, 3 warnings in 11.13s =====

```

Gambar 4 Hasil Test dari File yang ada di Branch Testing

link github : <https://github.com/Andre-Dir/Password-Hashing>